

Escape Routes: Airbnb Listings Analysis

ESCAPE ROUTES:

DATA WRANGLING AND ANALYSIS

OF AIRBNB LISTINGS ACROSS

U.S. CITIES

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Abstract

This study conducts a comprehensive empirical analysis of Airbnb's Open Listings dataset to examine the determinants of pricing, customer satisfaction, and host performance across major U.S. cities. Drawing upon the principles of data wrangling and business analytics, the research emphasizes the importance of transforming raw, heterogeneous data into structured, reliable information capable of supporting managerial and strategic decision-making. The dataset, obtained from Kaggle, comprises over 74,000 listings and 29 variables encompassing property characteristics, host information, amenities, and customer interaction metrics.

The methodological framework integrates key elements of the Extract-Transform-Load (ETL) process and exploratory data analysis (EDA). Tableau was utilized as the principal analytical platform for data cleansing, transformation, and visualization. During the wrangling phase, procedures such as null-value imputation, type conversion, deduplication, normalization, and outlier reduction (via 5th–95th percentile Winsorization) were applied to enhance data quality and interpretability. The transformed dataset was subsequently analyzed using Tableau dashboards to reveal spatial, categorical, and behavioral patterns in the Airbnb marketplace.

Findings indicate that pricing variation is primarily driven by location (city and neighborhood), room type, and amenity offerings. Entire homes and apartments consistently command the highest average nightly rates, whereas private and shared rooms appeal to price-sensitive segments. Amenities such as Wi-Fi, kitchen, parking, and air conditioning exhibit strong positive correlations with both price and review scores, underscoring the role of comfort features in value perception. Moreover, host responsiveness, verification, and super host status are significant predictors of customer satisfaction, as reflected in review ratings and booking frequency.

From a managerial perspective, the study contributes to the literature on data-driven hospitality management by demonstrating how systematic data wrangling and visualization can generate actionable insights for pricing optimization, reputation management, and competitive benchmarking. The research also provides implications for platform governance, emphasizing algorithmic transparency and data-based trust mechanisms in peer-to-peer economies. Ultimately, this study underscores the transformative potential of data wrangling as an analytical foundation for enhancing business intelligence, consumer understanding, and policy formulation within the digital accommodation ecosystem.

Keywords:

Airbnb, Data Wrangling, Pricing Analysis, Tableau, Customer Satisfaction, Data Visualization, Business Analytics

I. Introduction / Background

1.1 Overview

The rise of Airbnb has transformed the global hospitality landscape by offering travelers unique, localized experiences and allowing property owners to monetize their spaces. With millions of listings worldwide, understanding pricing strategies, customer satisfaction, and demand behavior has become critical to maintaining competitiveness.

1.2 Objective

This project aims to apply data wrangling and business analytics techniques to Airbnb's open dataset to:

- Identify key determinants of listing prices across major U.S. cities.
- Explore guest capacity patterns by room type.
- Analyze how cancellation policies and amenities affect pricing.
- Evaluate review scores as indicators of customer satisfaction.
- Provide recommendations for hosts and Airbnb to enhance profitability and user experience.

1.3 Research Questions

- How do average Airbnb prices differ across U.S. cities?
- What is the relationship between room type and guest capacity?
- How do amenities and cancellation policies influence price?

- What factors most strongly predict customer satisfaction?
- How can Airbnb hosts use these insights to optimize their listings?

1.4 Motivation

Airbnb's pricing and review systems are complex, driven by a combination of host characteristics, property attributes, and traveler preferences. Understanding these drivers helps improve transparency and promotes equitable market behavior. From a business analytics perspective, this dataset offers a rich opportunity to apply data wrangling, cleaning, and visualization skills in a real-world scenario.

1.5 Background of the Dataset

The Airbnb Open Listings dataset, sourced from Kaggle, contains 74,111 listings and 29 variables across multiple U.S. cities. Variables include property type, room type, host verification status, location (latitude, longitude, and neighborhood), price, review scores, and amenities. The dataset offers both structured (numeric, categorical) and semi-structured data (amenities text fields).

1.6 Report Structure

The report is structured as follows:

- Section 2: Data / Problem Analytics – data description, cleaning, and wrangling steps.
- Section 3: Discussion – interpretation of key analytical results.
- Section 4: Implications – practical applications and managerial insights.
- Section 5: Conclusion – summary of findings.
- Section 6: References.
- Section 7: Appendix – weekly schedule.

2. Data / Problem Analytics

2.1 Data Description

Data Source: Kaggle – Airbnb Open Listings Dataset

Dataset Size: 74,111 listings × 29 variables

Cities Covered: New York, Los Angeles, Chicago, San Francisco, Austin, and others.

Data Fields:

Category	Examples	Description
Property Information	property_type, room_type	Defines lodging category (Entire Home, Private Room, etc.)
Location	city, neighbourhood, latitude, longitude	Enables spatial mapping and price clustering
Pricing & Demand	price, minimum_nights, availability_365	Determines cost and occupancy trends
Customer Interaction	number_of_reviews, review_scores_rating	Reflects satisfaction and engagement
Host Characteristics	host_name, host_is_superhost, response_rate	Affects trust and booking probability
Amenities	wifi, kitchen, parking, pool, air_conditioning	Non-price factors influencing desirability

2.2 Methods

The methodology combines data wrangling and exploratory data analysis (EDA) using Tableau to uncover trends and relationships.

Step 1: Data Cleaning and Standardization

- Removed missing or inconsistent values (e.g., null prices or missing neighborhoods).
- Standardized price fields by removing currency symbols and converting text to numeric.
- Addressed duplicates based on host_id, property_name, and latitude/longitude.

Step 2: Data Transformation

- Created calculated fields (e.g., average price per room type).
- Normalized review scores to a 0–100 scale for comparison.
- Filtered out extreme outliers using Winsorization (5th and 95th percentiles).

Step 3: Data Visualization and Analysis

- Used Tableau’s mapping features to display geographical patterns of price and availability.
- Built interactive dashboards for:
 - Price by city and room type
 - Distribution of review scores

- Amenity frequency and impact on price
- Host responsiveness and customer satisfaction

Tableau Workflow: Data Wrangling

2.3 Data Analytics and Visualization

This section presents descriptive statistics and visualization results derived from the Airbnb dataset. Various Tableau dashboards were created to explore pricing trends, customer satisfaction patterns, amenity influence, and the impact of policies on listing performance.

2.3.1 Descriptive Statistics

Metric	Mean	Median	Standard Deviation	Range
log_price	4.782069	4.70953	0.717393785	7.600402
accommodates	3.155146	2	2.153588963	15
number_of_reviews	20.90057	6	37.8286413	605
review_scores_rating	72.84252	94	39.92050714	100

2.3.2 Visualization Results

A. Price Variation Across Cities

- Average prices highest in Los Angeles and New York.
- Lower prices in Boston and Chicago, reflecting local market conditions.

Insight: The bar chart demonstrates clear spatial variations in pricing across major U.S. cities. New York and Los Angeles display the highest average prices, reflecting higher living costs and tourism intensity. Boston and Chicago exhibit moderate price ranges, highlighting affordability in comparison to coastal cities. From a business perspective, hosts operating in competitive metropolitan markets should align prices strategically based on neighborhood demand patterns. Airbnb may use this information to refine dynamic pricing algorithms that respond to seasonal demand.

B. Price by Room Type

- Entire homes/apartments dominate listings (≈45%) with the highest average price.
- Private rooms (≈28%) show moderate pricing.
- Shared rooms have the lowest prices.

Insight: This visualization highlights the relationship between room type and average nightly rate. Entire homes and apartments, typically offering privacy and exclusivity, command premium prices. Private rooms cater to solo travelers and budget-conscious users, generating steady bookings at moderate rates. Shared rooms remain the most economical option, appealing to cost-sensitive guests. From a managerial standpoint, this differentiation underscores the importance of identifying target market segments and adjusting pricing to maximize occupancy and profitability.

C. Customer Satisfaction Trends

- Review scores positively correlate with host response rates and superhost status.
- Listings with more than 50 reviews have higher reliability (less variance in score).

Insight: The distribution of review scores reveals that entire homes and private rooms receive consistently high satisfaction ratings, largely due to better amenities and host responsiveness. Shared accommodations display greater variability, suggesting uneven quality experiences. Maintaining standardized cleanliness, communication, and safety protocols can help improve guest satisfaction across all listing types. For Airbnb, these findings indicate that enhanced review verification mechanisms could improve platform credibility.

D. Cancellation Policy Impact

- "Strict" cancellation policies correspond with slightly lower bookings and higher nightly rates.
- "Flexible" policies attract more guests but at lower prices.

Insight: Listings with 'Strict' cancellation policies tend to achieve higher prices but lower booking volumes, indicating a risk–return tradeoff for hosts. Flexible policies attract higher occupancy at lower nightly rates. Hosts should evaluate their cash flow preferences—prioritizing either consistent bookings or higher margins. Airbnb could optimize its interface by recommending policy-pricing pairings that match host objectives.

E. Trends in Review Scores by Room Type Over Time

- Review scores declined across all room types from 2008–2017.
- Entire homes surpassed private rooms in satisfaction after 2012.
- Shared rooms consistently received the lowest ratings.
- Hosts should improve amenities and responsiveness to boost scores.
- Airbnb can use review trends to guide new hosts and promote quality listings.

Insight: Between 2008 and 2017, average review scores for Airbnb listings declined across all room types, reflecting rising guest expectations and market saturation. Entire homes overtook

private rooms in satisfaction around 2012, maintaining higher ratings until 2016, while shared rooms consistently lagged. These trends suggest that hosts must prioritize quality, responsiveness, and amenities to remain competitive. For Airbnb, leveraging historical review data can help improve host onboarding, highlight reliable listings, and enhance overall platform trust.

F. Guest Capacity by Property Type and Booking Flexibility

- Vacation homes and villas host the most guests.
- Instant bookable listings tend to accommodate more people.
- Unique stays (e.g., yurts, treehouses) suit smaller groups.
- Hosts should enable instant booking for larger properties.
- Airbnb can promote high-capacity listings for group travel.

Insight: The chart shows that vacation homes, villas, and townhouses accommodate the most guests, making them ideal for families or group travelers. Listings with instant booking options generally support more guests, suggesting that booking flexibility enhances appeal. Niche properties like yurts and treehouses cater to smaller, experience-driven segments. Hosts can boost visibility by enabling instant booking, while Airbnb can optimize search filters to highlight spacious listings during peak seasons.

G. Key Business Insights from Property Review Scores

- Top-rated: Castles, Islands, Treehouses score near 100.
- Low-rated: Timeshares, Lofts score around 50.
- Mid-range: Most properties score between 60–80.
- Trend: Unique stays attract better reviews.
- Opportunity: Invest in experiential properties.

Insight: The bar chart reveals a clear trend in customer preferences within the hospitality industry: unique and unconventional property types such as castles, islands, and treehouses consistently receive the highest average review scores, suggesting that travelers value distinctive and memorable experiences. In contrast, more traditional accommodations like timeshares and lofts tend to receive lower ratings, indicating a potential decline in their appeal. Most property types of cluster around moderate review scores, but the standout performance of niche properties presents a strategic opportunity for businesses to differentiate themselves. By investing in or promoting unique lodging options, hospitality providers can enhance customer satisfaction and potentially increase bookings.

3. Discussion

The comprehensive analysis of Airbnb listings across major U.S. cities reveals significant spatial and categorical influences on pricing and customer satisfaction. Eight key dimensions emerged from the data visualizations:

- **City-Level Pricing Dynamics** Listings in metropolitan hubs such as New York and San Francisco consistently command higher nightly rates. This trend reflects the influence of urban density, tourism demand, and local economic conditions. Hosts in these cities can leverage this insight to competitively price their listings while considering neighborhood-specific demand patterns.
- **Room Type Preferences and Pricing** Entire homes and apartments are the most sought-after accommodations, particularly among families and groups seeking privacy and comfort. These listings attract premium pricing. In contrast, private and shared rooms cater to budget-conscious travelers, offering more affordable options with moderate review scores.
- **Host Engagement and Service Quality** Host responsiveness and verified status are strongly correlated with higher review scores. Super hosts receive approximately 12% higher ratings on average. This underscores the importance of timely communication, reliability, and professionalism in enhancing guest satisfaction and fostering repeat bookings.
- **Customer Satisfaction and Review Consistency** Listings with a higher number of reviews tend to exhibit more consistent ratings, suggesting that volume contributes to trust and reliability. Variability in review scores decreases as the number of reviews increases, indicating that well-reviewed properties are perceived as more dependable by prospective guests.
- **Cancellation Policy Trade-offs** Listings with strict cancellation policies tend to have higher nightly rates but lower booking volumes. Conversely, flexible policies attract more guests, albeit at reduced prices. This trade-off highlights the need for hosts to align their cancellation terms with their revenue and occupancy goals.
- **Temporal Trends in Review Scores** Between 2008 and 2017, review scores declined across all room types, reflecting rising guest expectations and increased competition. Entire homes surpassed private rooms in satisfaction after 2012, maintaining higher ratings through 2016. These trends suggest that hosts must continuously improve service quality to remain competitive.
- **Guest Capacity and Booking Flexibility** Larger property types such as vacation homes and villas accommodate more guests and are often enabled with instant booking features. This combination enhances appeal for group travelers and families. Unique stays like yurts and treehouses, while hosting fewer guests, cater to niche markets seeking experiential travel.

- **Property Type and Review Scores** Unique accommodations—such as castles, islands, and treehouses—consistently receive the highest review scores, indicating a strong preference for distinctive and memorable experiences. Traditional property types like timeshares and lofts tend to receive lower ratings, suggesting a need for innovation or repositioning.

The data wrangling process, including null-value imputation, normalization, and outlier reduction, was instrumental in ensuring the accuracy and reliability of these insights. By transforming raw data into structured formats, the analysis provided a robust foundation for strategic decision-making in the hospitality sector.

4. Implications

4.1 Managerial Implications for Hosts

- **Pricing Optimization:** Hosts should utilize city-level and room-type data to benchmark competitive pricing strategies, ensuring alignment with local demand and seasonal trends.
- **Targeted Marketing:** Promoting unique property types such as castles, islands, and treehouses can attract experience-driven travelers and command premium pricing.
- **Booking Strategy:** Hosts must evaluate the trade-offs between strict and flexible cancellation policies to balance occupancy rates and revenue goals.
- **Review Management:** Encouraging positive guest reviews through responsive communication and consistent service quality can enhance listing credibility and visibility.
- **Capacity Planning:** Hosts should optimize guest capacity based on property type and consider enabling instant booking for larger accommodations to appeal to group travelers.

4.2 Implications for the Airbnb Platform

- **Algorithm Enhancement:** Airbnb can refine its pricing algorithms by incorporating host responsiveness and cancellation policy data to better match guest preferences.
- **Quality Assurance:** Elevating the visibility of verified and superhost listings can improve platform trust and encourage higher service standards.
- **Data Transparency:** Providing clearer breakdowns of pricing determinants and review metrics can empower users to make informed decisions and foster trust.

4.3 Implications for Policy Makers and Tourism Boards

- **Urban Planning and Zoning:** Insights into spatial concentration of listings can inform housing affordability strategies and guide zoning regulations to balance residential and tourism needs.

- **Tourism Infrastructure Development:** Understanding demand patterns across cities and property types can support targeted investments in tourism infrastructure and services.

4.4 Implications for Customers

- **Informed Decision-Making:** Greater awareness of factors influencing pricing and satisfaction enables travelers to select accommodations that best meet their needs and expectations.
- **Trust and Safety:** Review volume and host verification serve as reliable indicators of listing quality, helping guests make safer and more satisfying booking choices.

Managerial Decision Framework Derived from Analysis

5. Conclusion

This study applied data wrangling and visualization techniques to analyze Airbnb listings across major U.S. cities, revealing key insights from eight distinct graphs:

1. **Price Variation Across Cities:** Listings in cities like New York and Los Angeles command higher prices due to tourism and urban density.

2. **Price by Room Type:** Entire homes and apartments are priced higher, while shared rooms remain budget friendly.s

3. **Customer Satisfaction Trends:** Review scores correlate with host responsiveness and superhost status.

4. **Cancellation Policy Impact:** Strict policies yield higher prices but fewer bookings; flexible policies attract more guests.

5. **Review Score Trends Over Time:** Satisfaction declined across all room types from 2008–2017, with entire homes outperforming others.

6. Guest Capacity by Property Type: Vacation homes and villas host larger groups; instant booking enhances capacity.

7. Unique Property Ratings: Castles, islands, and treehouses receive top reviews, indicating demand for experiential stays.

8. Business Insights: Unique accommodations outperform traditional ones, suggesting strategic investment opportunities.

These findings provide actionable guidance for hosts to refine pricing, improve guest experience, and differentiate their offerings. Airbnb and policymakers can leverage these insights to enhance platform governance, support sustainable tourism, and foster innovation in the sharing economy.

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- Appendix – Project Schedule

Week	Task	Description
Week 1	Topic Selection & Proposal	Finalize research question and submit project proposal
	Data Collection	Obtain Airbnb dataset from Kaggle

Week	Task	Description
	Data Cleaning	Handle missing values, duplicate, and standardize formats
Week 2	Data Transformation	Normalize variables, create calculated fields
	Exploratory Data Analysis	Generate summary statistics and initial Tableau visuals
	Visualization Development	Build dashboards for pricing, satisfaction, and amenities
Week 3	Interpretation	Discuss insights, patterns, and implications
	Report Writing	Draft and finalize full academic report
	Presentation Prep	Create PowerPoint presentation and finalize deliverables